

3.4 HVAC Rough-In Guide

System design, duct layout and equipment rough-in recorded room by room, with the load calculation, quality checklist and inspection result.

Property Address: _____

Permit Number: _____ Inspection Date: _____

HVAC Contractor: _____ License #: _____

Phone: _____

INSTRUCTIONS: Fill in the system overview and load calculation before any duct goes up — undersized returns and guessed equipment capacity are the two most common and most expensive HVAC mistakes an owner-builder makes. Record actual duct and register sizes as installed.

SYSTEM OVERVIEW

Equipment & Controls Summary

System Component	Type / Specification	Notes
Heating System Type	☐ Forced Air ☐ Radiant ☐ Heat Pump	
Heating Source	☐ Gas ☐ Electric ☐ Oil ☐ Heat Pump	
Cooling System Type	☐ Central AC ☐ Heat Pump ☐ Mini-Split	
System Capacity — Heating	BTU: _____	
System Capacity — Cooling	Tons: _____	
Furnace/Air Handler Location	☐ Basement ☐ Attic ☐ Closet ☐ Garage	
Condensing Unit Location	Location: _____	
Zoned System	☐ Yes ☐ No If yes, zones: _____	

Equipment & Controls Summary

System Component	Type / Specification	Notes
Thermostat Type	☐ Programmable ☐ Smart ☐ Manual	
Number of Thermostats	Qty: _____	

LOAD CALCULATION

MANUAL J LOAD CALCULATION PERFORMED: ☐ Yes ☐ No

Calculated By: _____ Date: _____

Total Heating Load (BTU): _____

Total Cooling Load (BTU): _____ or Tons: _____

LOAD CALCULATION DOCUMENTS ATTACHED: ☐ Yes ☐ No

DUCTWORK LAYOUT — SUPPLY

Supply Registers & Branch Ducts

	Location / Room	Register Size	CFM Req'd	Duct Size	Notes
☐	Main trunk line	N/A			
☐	Master Bedroom				
☐	Master Bathroom				
☐	Bedroom #2				
☐	Bedroom #3				
☐	Bedroom #4				
☐	Bathroom #2				
☐	Living Room				

Supply Registers & Branch Ducts

	Location / Room	Register Size	CFM Req'd	Duct Size	Notes
☐	Dining Room				
☐	Kitchen				
☐	Family Room				
☐	Office/Den				
☐	Hallway(s)				
☐	Basement/Lower Level				
☐	Bonus Room				
☐	Other: _____				

Supply Ductwork Notes

DUCTWORK LAYOUT — RETURN AIR

Return Grilles & Return Ducts

	Location	Return Grille Size	Duct Size	Notes
☐	Main return trunk	N/A		
☐	Central hallway return			
☐	Master bedroom return			
☐	Bedroom #2 return			

Return Grilles & Return Ducts

	Location	Return Grille Size	Duct Size	Notes
☐	Living area return			
☐	Basement return			
☐	Second floor return			
☐	Other: _____			

Total Return Air CFM Required: _____

RETURN AIR ADEQUATELY SIZED FOR SYSTEM: ☐ Yes ☐ No

RETURN AIR LOCATED AWAY FROM SUPPLY REGISTERS: ☐ Yes ☐ No

Return Air Sizing Notes

EQUIPMENT LOCATIONS

Equipment & Thermostat Placement

	Equipment / Item	Location	Notes / Requirement
☐	Furnace/Air Handler		
☐	Equipment clearances verified		Per manufacturer specs
☐	Access for service/filter change		Adequate access provided
☐	Condensing unit pad location		
☐	Condensing unit clearances		Min. 24" one side, 12" others
☐	Thermostat location — Zone 1		

Equipment & Thermostat Placement

	Equipment / Item	Location	Notes / Requirement
☐	Thermostat location — Zone 2		
☐	Thermostat on interior wall		Not on exterior wall
☐	Thermostat away from heat sources		Not near windows, vents, etc.
☐	Zone dampers (if applicable)		

UTILITIES & CONNECTIONS REQUIRED

Gas Line (if applicable)

	Item	Specification	Notes
☐	Gas line size to furnace	Per BTU load/distance	
☐	Gas shut-off valve at equipment	Within 6' of unit	
☐	Gas line properly supported	Per code	
☐	Sediment trap installed	Drip leg at appliance	
☐	Gas line pressure tested	☐ Yes ☐ Pending	

Electrical Requirements

	Item	Specification	Notes
☐	Furnace/Air Handler circuit	Amps (dedicated): ____	
☐	Condensing unit circuit	Amps (dedicated): ____	
☐	Disconnect switch at condensing unit	Within sight of unit	
☐	Thermostat wiring — 18/5 or greater	Common wire included	
☐	Zone control wiring (if applicable)	Per system specs	

Condensate Drain

	Item	Specification	Notes
☐	Primary condensate drain line	3/4" PVC typical	
☐	Drain line properly sloped	1/4" per foot minimum	
☐	Drain terminates to approved location	Floor drain, exterior, etc.	
☐	P-trap in condensate line	Required for proper operation	
☐	Secondary/emergency drain pan (if req.)	For attic installations	
☐	Emergency drain visible termination	Alert to primary drain clog	

Combustion Air & Venting (Gas/Oil Systems)

	Item	Specification	Notes
☐	Combustion air provisions adequate	Per manufacturer/code	
☐	Vent pipe size correct	Per appliance specs	
☐	Vent termination location approved	Clearances from openings	
☐	Vent properly supported/sloped	Per manufacturer	

ROUGH-IN QUALITY CHECKLIST

Ductwork Installation

	Item	Notes
☐	All ductwork installed per approved plan	
☐	Duct sizes match design specifications	
☐	Ductwork properly supported every 4'-6'	
☐	All joints sealed with mastic or approved tape	
☐	No cloth duct tape used (not code compliant)	

Ductwork Installation

	Item	Notes
☐	Flexible duct fully extended (no sagging)	
☐	Flexible duct runs limited to 8'-10' max	
☐	Proper transition fittings used	
☐	Return air plenums sealed at equipment	
☐	No return air leaks in unconditioned spaces	
☐	Supply boots properly installed and sealed	
☐	Boot supports installed in ceiling/floor	
☐	Register/grille locations marked	
☐	Ductwork in unconditioned spaces insulated	
☐	Insulation R-value meets code minimum	
☐	Vapor barrier on insulation facing outward	
☐	No sharp bends or kinks in ductwork	
☐	Turning vanes in square elbows (if required)	

Ductwork Notes

Equipment Rough-In

	Item	Notes
☐	Equipment platform/stand level and secure	
☐	Vibration isolation if required	
☐	Refrigerant line set path planned/installed	

Equipment Rough-In

	Item	Notes
☐	Line set properly sized for system	
☐	Line set insulated and protected	
☐	Line set penetrations sealed	
☐	Thermostat wire run to all locations	
☐	Extra thermostat wire coiled at equipment	
☐	Condensate drain rough-in complete	
☐	Equipment access door/panel clearance verified	

Penetrations & Sealing

	Item	Notes
☐	All duct penetrations through fire-rated walls sealed	
☐	Duct penetrations through roof properly flashed	
☐	Duct penetrations to exterior sealed airtight	
☐	No ducts running through garage (if code prohibits)	
☐	Fire dampers installed where required	
☐	Backdraft dampers on exhaust vents	

Code Compliance

	Item	Notes
☐	System meets energy code requirements	
☐	Equipment SEER/AFUE ratings meet minimums	
☐	Duct insulation meets energy code	
☐	Combustion air meets code requirements	

Code Compliance

	Item	Notes
☐	Equipment clearances meet manufacturer specs	
☐	Gas venting meets code and manufacturer specs	
☐	Return air not from garage or bathrooms	

ADDITIONAL HVAC COMPONENTS

Ventilation & Exhaust

	Item / Location	Specification	Notes
☐	Whole-house ventilation system	☐ ERV ☐ HRV ☐ Exhaust ☐ N/A	
☐	Kitchen range hood exhaust duct	To exterior CFM: _____	
☐	Bathroom exhaust — Master	To exterior CFM: _____	
☐	Bathroom exhaust — Bath #2	To exterior CFM: _____	
☐	Bathroom exhaust — Bath #3	To exterior CFM: _____	
☐	Laundry room exhaust (if req.)	To exterior	
☐	Dryer vent duct	4" rigid to exterior	
☐	Dryer vent length within limits	Max length per code	
☐	All exhaust ducts terminate outside	Not into attic/crawlspace	
☐	Exhaust vent terminations have caps	Pest/weather protection	

Air Quality & Filtration

	Item	Specification	Notes
☐	Filter location/access planned		
☐	Filter size W: _____ H: _____ D: _____		
☐	Upgraded filtration system (if applicable)	☐ HEPA ☐ Electronic ☐ Media ☐ Standard	
☐	UV light system (if applicable)	Location: _____	
☐	Humidifier rough-in (if applicable)	☐ Whole-house ☐ Bypass ☐ Steam	
☐	Humidifier water line to equipment	1/4" line with shutoff	
☐	Dehumidifier provisions (if applicable)		

General Notes

INSPECTION RESULTS & CORRECTIONS

Initial HVAC Rough-In Inspection

Inspection Date: _____ Time: _____

Inspector Name: _____ Inspector Badge/ID: _____

RESULT: ☐ PASSED ☐ FAILED — Corrections Required

Deficiencies / Code Violations Found

Corrections Made

Date Corrections Completed: _____

HVAC Contractor: _____ License #: _____

Description of Corrections

Re-Inspection (if required)

Re-Inspection Date: _____ Time: _____

Inspector Name: _____

RESULT: ☐ PASSED ☐ FAILED — Additional Corrections Required**Additional Comments**

FINAL SIGN-OFF

Owner/Builder Signature: _____ Date: _____

HVAC Contractor Signature: _____ Date: _____

Inspector Signature: _____ Date: _____

Permit Number: _____

FINAL APPROVAL: ☐ YES